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Exercise 2

Assume for contradiction that there exists $a, b \in \mathbb{Z}$ with $b \neq 0$ such that $\left(\frac{a}{b}\right)^2 = 2$. If a is negative and b is positive, then replacing a with $-a$ gives us $\left(\frac{-a}{b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $-a, b$ is also a valid pair of integers satisfying the conditions. If b is negative and a is positive, then replacing b with $-b$ gives us $\left(\frac{a}{-b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $a, -b$ is also a valid pair of integers satisfying the conditions. If both a and b are negative, then replacing a with $-a$ and b with $-b$ gives us $\left(\frac{-a}{-b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $-a, -b$ is also a valid pair of integers satisfying the conditions. Hence, we can assume without loss of generality that $a, b > 0$.

Since we assume that b only takes values in $\mathbb{N}$, by the well-ordering principle, there exists a smallest b satisfying the conditions. Note that

$$\begin{aligned}
 \left(\frac{2b-a}{a-b}\right)^2 &= \frac{4b^2 - 4ab + a^2}{a^2 - 2ab + b^2} \\
 &= \frac{4 - 4\left(\frac{a}{b}\right) + \left(\frac{a}{b}\right)^2}{\left(\frac{a}{b}\right)^2 - 2\left(\frac{a}{b}\right) + 1} \\
 &= \frac{4 - 4\left(\frac{a}{b}\right) + 2}{2 - 2\left(\frac{a}{b}\right) + 1} \\
 &= \frac{6 - 4\left(\frac{a}{b}\right)}{3 - 2\left(\frac{a}{b}\right)} \\
 &= \frac{6 - 4\left(\frac{a}{b}\right)}{3 - 2\left(\frac{a}{b}\right)} \times \frac{3 + 2\left(\frac{a}{b}\right)}{3 + 2\left(\frac{a}{b}\right)} \\
 &= \frac{18 + 12\left(\frac{a}{b}\right) - 12\left(\frac{a}{b}\right) - 8\left(\frac{a}{b}\right)^2}{9 - 4\left(\frac{a}{b}\right)^2} \\
 &= \frac{18 - 8 \times 2}{9 - 4 \times 2} \\
 &= 2.
 \end{aligned}$$

Recall that $\left(\frac{a}{b}\right)^2 = 2$, so we have $a^2 = 2b^2$ and thus $2b^2 - a^2 = 0$. Note that

$$\begin{aligned}
 (2b-a)(2b+a) &= 4b^2 - a^2 \\
 &= 2b^2 + (2b^2 - a^2) \\
 &= 2b^2 \\
 &> 0
 \end{aligned}$$

since $b > 0$ by assumption. Since $a, b > 0$, we have $2b + a > 0$. Hence, $2b - a > 0$. Also, from $a^2 = 2b^2$ we have $a^2 - b^2 = b^2$. Considering $b > 0$, we have

$$\begin{aligned} b^2 &> 0 \\ a^2 - b^2 &> 0 \\ (a - b)(a + b) &> 0 \\ a - b &> 0 \end{aligned}$$

since $a + b > 0$. From $2b - a > 0$ and $a - b > 0$ we know that $2b - a, a - b$ is also a pair of positive integers satisfying the conditions. However, from $2b - a > 0$ we have $a < 2b$, and hence $a - b < 2b - b = b$, contradicting the minimality of b . $\square$

Exercise 5

Let $S = \{p + q\alpha + r\alpha^2 \mid p, q, r \in \mathbb{Q}\}$. We first prove that S is a subring of $\mathbb{C}$.

Note that $1 = 1 + 0\alpha + 0\alpha^2 \in S$. Let $p + q\alpha + r\alpha^2, p' + q'\alpha + r'\alpha^2 \in S$, where $p, q, r, p', q', r' \in \mathbb{Q}$. Then we have $(p + q\alpha + r\alpha^2) + (p' + q'\alpha + r'\alpha^2) = (p + p') + (q + q')\alpha + (r + r')\alpha^2 \in S$, $-(p + q\alpha + r\alpha^2) = (-p) + (-q)\alpha + (-r)\alpha^2 \in S$ and $(p + q\alpha + r\alpha^2)(p' + q'\alpha + r'\alpha^2) = (pp' + 2rq') + (pq' + p'q + 2rr')\alpha + (pr' + qq' + qr' + rp')\alpha^2 \in S$. By the subring test, S is a ring.

Next, we construct an inverse explicitly to show that S is a subfield. Let $x = p + q\alpha + r\alpha^2 \in S \setminus \{0\}$, where $p, q, r \in \mathbb{Q}$. Then we have $x^2 = (p^2 + 4qr) + (2pq + 2r^2)\alpha + (2pr + q^2)\alpha^2$ and $x^3 = (p^3 + 12pqr + 2q^3 + 4r^3) + (3p^2q + 6pr^2 + 6q^2r)\alpha + (3p^2r + 3pq^2 + 6qr^2)\alpha^2$. Let $x^3 + k_2x^2 + k_1x + k_0 = 0$. Denote the coefficient of α^j in x^i by x_{ij} . Then we have $(x_{30} + x_{31}\alpha + x_{32}\alpha^2) + k_2(x_{20} + x_{21}\alpha + x_{22}\alpha^2) + k_1(p + q\alpha + r\alpha^2) + k_0 = 0$, that is, $(x_{30} + x_{20}k_2 + pk_1 + k_0) + (x_{31} + x_{21}k_2 + qk_1)\alpha + (x_{32} + x_{22}k_2 + rk_1)\alpha^2 = 0$. Since $1, \alpha, \alpha^2$ are linearly independent, comparing coefficients gives us the linear system

$$\begin{cases} k_0 + pk_1 + x_{20}k_2 = -x_{30} \\ qk_1 + x_{21}k_2 = -x_{31} \\ rk_1 + x_{22}k_2 = -x_{32} \end{cases}.$$

We can use the following python code to solve this system.

```

1  import sympy as sp
2
3  k0, k1, k2 = sp.symbols('k0 k1 k2')
4  p, q, r = sp.symbols('p q r')
5
6  x30 = p**3 + 12 * p * q * r + 2 * (q**3) + 4 * (r**3)
7  x31 = 3 * (p**2) * q + 6 * p * (r**2) + 6 * (q**2) * r
8  x32 = 3 * (p**2) * r + 3 * p * (q**2) + 6 * q * (r**2)
9  x20 = p**2 + 4 * q * r
10 x21 = 2 * p * q + 2 * (r**2)
11 x22 = 2 * p * r + q**2
12
13 M = sp.Matrix([
14     [1, p, x20, - x30],
15     [0, q, x21, - x31],
16     [0, r, x22, - x32]
17 ])

```

```

18
19 solution = sp.solve_linear_system(M, k0, k1, k2)
20
21 for var in [k0, k1, k2]:
22     print(f"{var} =")
23     sp.pprint(sp.simplify(solution[var]))

```

The solution is $k_0 = -p^3 + 6pqr - 2q^3 - 4r^3$, $k_1 = 3p^2 - 6qr$ and $k_2 = -3p$. Since $x^3 + k_2x^2 + k_1x + k_0 = 0$, we have $x(x^2 + k_2x + k_1 + k_0x^{-1}) = 0$. Recall that $\mathbb{C}$ is a field and hence a domain, and $x \neq 0$. Therefore, $x^2 + k_2x + k_1 + k_0x^{-1} = 0$, that is,

$$x^{-1} = -\frac{1}{k_0}x^2 - \frac{k_2}{k_0}x - \frac{k_1}{k_0}.$$

Since x and x^2 are linear combinations of $1, \alpha, \alpha^2$, x^{-1} is also a linear combination of $1, \alpha, \alpha^2$. □